# TabuLLM: AI Benchmark Report

## 1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

| Metric | Consolidated Value |
|---|---|
| Avg Fidelity | 0.85 |

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## Dataset Audit: demo_data.csv

### 1. Dataset Portrait

| Attribute | Value |
|---|---|
| Rows | 0 |
| Cols | 0 |

### 2. Generation Strategy & LLM Config [1]

| Parameter | Configuration |
|---|---|
| Algorithm (*12) | google/gemma-3-4b |

### 3. Statistical Significance & Data Drift

### 4. Quality & Fidelity Metrics (Mean ± Std)

| Metric Category | Aggregated Score |
|---|---|
| Fidelity (*1) | 0.85 |

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] Synthetic Fidelity (JSD): Overall distribution similarity.*

*[3] Correlation Stability: Multivariate relationships preserved.*

*[4] Distribution Consistency: Median feature similarity.*

*[5] Entropy Preservation: Information diversity maintained.*

*[6] Mean Z Score Drift: Standardized distribution shift.*
*[7] Categorical Fidelity: Category distribution similarity.*
*[8] Privacy Risk: Synthetic rows too close to originals.*
*[9] Data Density: Non-missing cells in augmented data.*
*[10] Coverage Score: Synthetic coverage of original space.*
*[11] Feature Accuracy: Mean feature similarity.*
*[12] Novelty Score: Distance ratio to nearest neighbors.*

## 6. Downstream ML Utility: Comparative Analysis

**Model: RF**

| Metric | Real (Baseline) | Augmented | Delta (Lift) |
|---|---|---|---|
| ACCURACY | 0.8 | 0.82 | 0.02 |
| F1 | 0.75 | 0.78 | 0.03 |

## Appendix: Benchmark Data Trace

*Legend: [Light Blue] = Original Context Rows | [Light Green] = Generated Synthetic Rows*

### A. Baseline: Original Dataset (Full Record Sample)